KARTHIK RAMKUMAR - U24AI001

Dimensionality Reduction Using Principal Component Analysis (PCA)

In [ ]:
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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

df = pd.read_csv('titanic.csv')
print("Original Dataset Shape:", df.shape)

df.columns = df.columns.str.lower()

numerical_features = ['age', 'fare', 'sibsp', 'parch', 'pclass']
df_num = df[numerical_features].copy()

df_num['age'] = df_num['age'].fillna(df_num['age'].median())
df_num['fare'] = df_num['fare'].fillna(df_num['fare'].median())

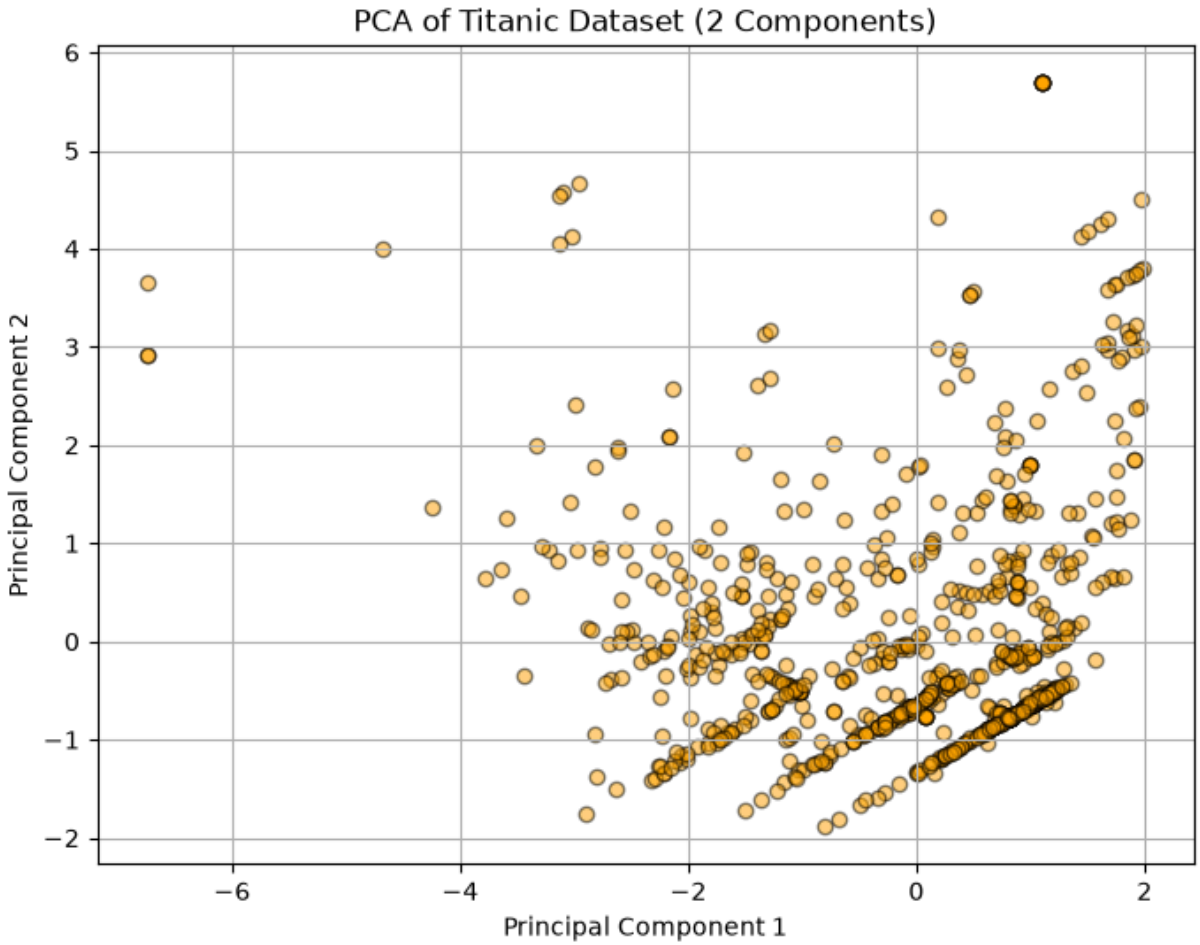
scaler = StandardScaler() #standardize the features so that their mean value
scaled_data = scaler.fit_transform(df_num)
# calculates the explained valence ratio( two components capture ~66.5% of c
pca = PCA(n_components=2)
pca_data = pca.fit_transform(scaled_data)

explained_variance = pca.explained_variance_ratio_
print(f"\nexplained variance ratio(2 components):{explained_variance}")
print(f"total variance explained:{sum(explained_variance)*100:.2f}%")

# visualize the first two principal components
plt.figure(figsize=(8, 6))
plt.scatter(pca_data[:, 0], pca_data[:, 1], alpha=0.5, c='orange', edgecolor
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.title('PCA of Titanic Dataset (2 Components)')
plt.grid(True)
plt.show()
```

```
Original Dataset Shape: (891, 15)

Explained Variance Ratio (2 components): [0.33960812 0.32520788]
Total Variance Explained: 66.48%
```

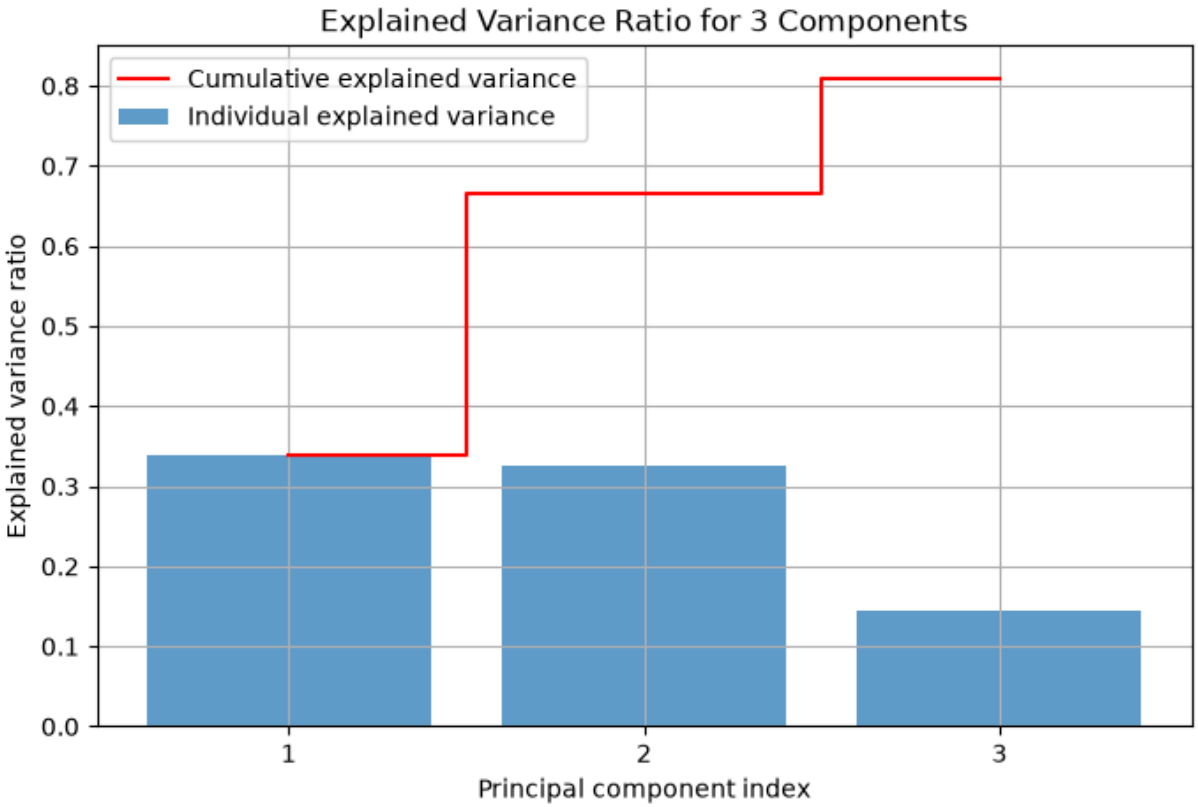


In [ ]:
```
# 1 & 2: Apply PCA using three principal components and plot explained varia
pca_3 = PCA(n_components=3)
pca_data_3 = pca_3.fit_transform(scaled_data)

explained_variance_3 = pca_3.explained_variance_ratio_
print(f"Explained Variance Ratio (3 components): {explained_variance_3}")
print(f"Total Variance Explained (3 components): {sum(explained_variance_3)

plt.figure(figsize=(8, 5))
plt.bar(range(1, 4), explained_variance_3, alpha=0.7, align='center', label=
plt.step(range(1, 4), np.cumsum(explained_variance_3), where='mid', label='C
plt.ylabel('Explained variance ratio')
plt.xlabel('Principal component index')
plt.title('Explained Variance Ratio for 3 Components')
plt.xticks([1, 2, 3])
plt.legend(loc='best')
plt.grid(True)
plt.show()
```

```
Explained Variance Ratio (3 components): [0.33960812 0.32520788 0.14457332]
Total Variance Explained (3 components): 80.94%
```



In [ ]:
```
# 3: Compare the dataset before and after PCA
print("Original Dataset Shape (Selected Features):", df_num.shape)
print("Transformed Dataset Shape (2 Components):", pca_data.shape)

pca_df = pd.DataFrame(data=pca_data, columns=['PC1', 'PC2'])
print("\nfirst 5 rows of original standardized data:\n", df_num.head())
print("\nfirst 5 rows of PCA transformed data (2 Components):\n", pca_df.hea
```

```
Original Dataset Shape (Selected Features): (891, 5)
Transformed Dataset Shape (2 Components): (891, 2)

First 5 rows of original standardized data:
     age     fare  sibsp  parch  pclass
0  22.0   7.2500      1      0       3
1  38.0  71.2833      1      0       1
2  26.0   7.9250      0      0       3
3  35.0  53.1000      1      0       1
4  35.0   8.0500      0      0       3

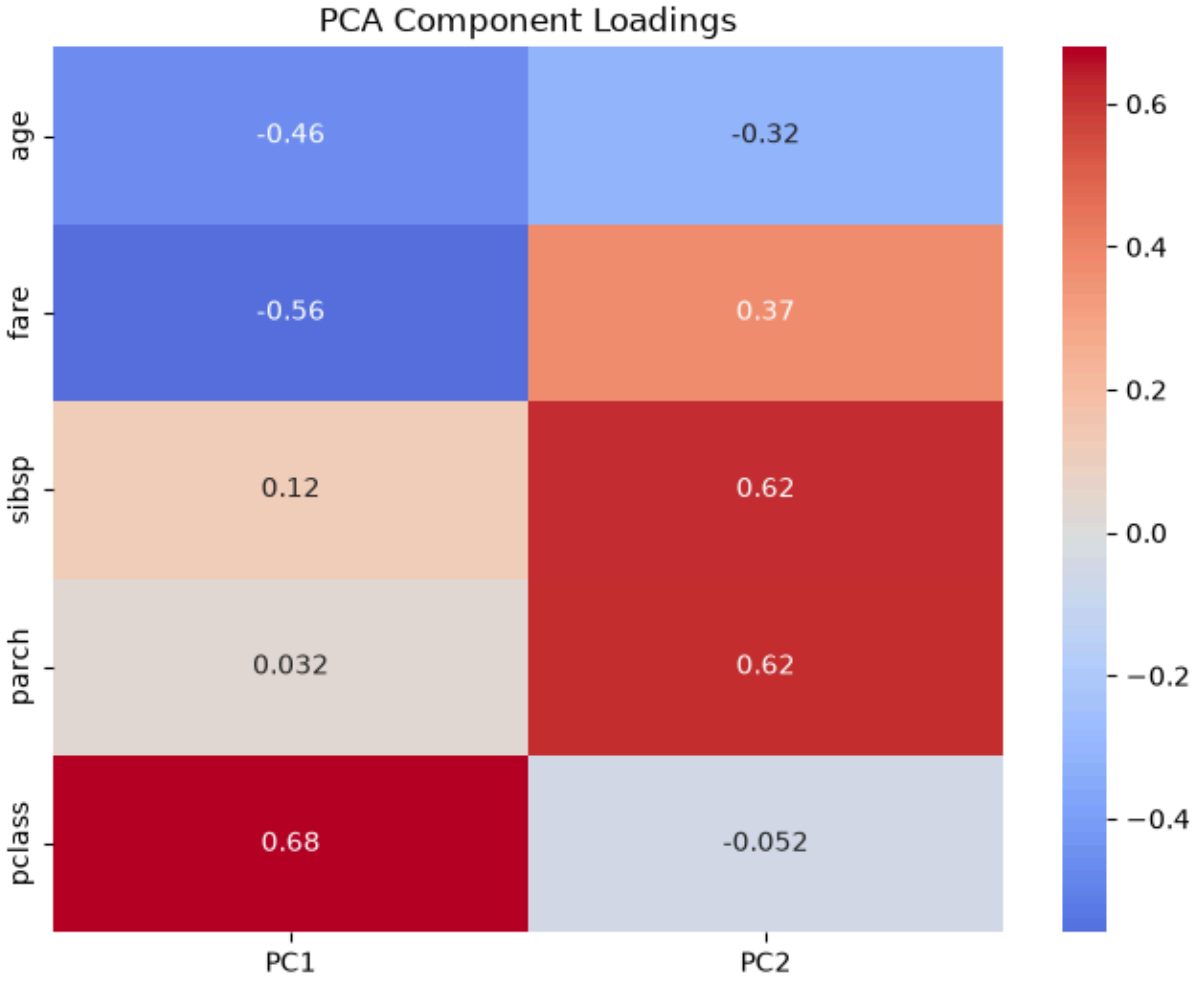
First 5 rows of PCA transformed data (2 Components):
         PC1       PC2
0  1.140660 -0.076520
1 -1.768445  0.133479
2  0.879196 -0.726606
3 -1.458103  0.071555
4  0.561252 -0.943713
```

In [ ]:
```
# 4: Interpret the first two principal components
loadings = pd.DataFrame(pca.components_.T, columns=['PC1', 'PC2'], index=num
print("PCA Loadings (Feature Contributions):\n", loadings)
```

```
plt.figure(figsize=(8, 6))
sns.heatmap(loadings, annot=True, cmap='coolwarm', center=0)
plt.title('PCA Component Loadings')
plt.show()

print("\nInterpretation:")
print("PC1 is typically influenced strongly pclass, fare and age majorly")
print("PC2 often captures variance related sibsb and parch")
```

```
PCA Loadings (Feature Contributions):
               PC1       PC2
age    -0.457655 -0.315236
fare   -0.559469  0.367648
sibsp   0.124736  0.615183
parch   0.031850  0.619955
pclass  0.678950 -0.051643
```




```
Interpretation:
PC1 is typically influenced strongly by passenger class and fare.
PC2 often captures variance related to family size (sibsp/parch) or age.
```